

lncRNAlyzr: Enrichment Analysis for lncRNA Sets ☆

John Erol Evangelista [1], Tahleel Ali-Nasser [2], Lauren E. Malek [1], Zhuorui Xie [1], Giacomo B. Marino [1], Assaf C. Bester [2], and Avi Ma'ayan [1,*]

*1 - **Mount Sinai Center for Bioinformatics,** Department of Pharmacological Sciences, Department of Artificial Intelligence and Human Health, Icahn School of Medicine at Mount Sinai, One Gustave L. Levy Place, Box 1603, New York, NY 10029, USA*
*2 - **Department of Biology,** Technion-Israel Institute of Technology, 3200003 Haifa, Israel*

***Correspondence to Avi Ma'ayan:** JohnErol.Evangelista@mssm.edu (J.E. Evangelista), tahlil-ali@campus.technion.ac.il (T. Ali-Nasser), lauren.malek@mssm.edu (L.E. Malek), zhuorui.xie@mssm.edu (Z. Xie), giacomo.marino@mssm.edu (G.B. Marino), bestera@technion.ac.il (A.C. Bester), avi.maayan@mssm.edu (A. Ma'ayan)*
https://doi.org/10.1016/j.jmb.2025.168938
Editor: David Mathews

Abstract

lncRNAs make up a large portion of the human genome affecting many biological processes in normal physiology and diseases. However, human lncRNAs are understudied compared to protein-coding genes. While there are many tools for performing gene set enrichment analysis for coding genes, few tools exist for lncRNA enrichment analysis. lncRNAlyzr is a webserver application designed for lncRNAs enrichment analysis. lncRNAlyzr has a database containing 33 lncRNA set libraries created by computing correlations between lncRNAs and annotated coding gene sets. After users submit a set of lncRNAs to lncRNAlyzr, the enrichment analysis results are visualized as ball-and-stick subnetworks where nodes are lncRNAs connected to enrichment terms from across selected lncRNA set libraries. To demonstrate lncRNAlyzr, it was used to analyze the effects of knocking down the lncRNA CYTOR in K562 cells. Overall, lncRNAlyzr is an enrichment analysis tool for lncRNAs aiming to further our understanding of lncRNAs functional modules. lncRNAlyzr is available from: https://lncrnalyzr.maayanlab.cloud.
© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Introduction

Non-coding RNAs with over 200 nucleotides are considered long non-coding RNAs (lncRNAs). Since proteins are vital for all cellular functions, and lncRNAs do not encode for proteins, lncRNAs were initially assumed to be less important than coding genes. However, there is now broad acceptance that lncRNAs play critical roles in many intracellular and extracellular molecular processes in normal physiology and in disease. For example, many lncRNAs have been implicated in regulating gene expression and were shown to be key drivers in various cancers.[1] While the broad functions of only a few lncRNAs are known, the roles of most human and mouse lncRNAs are not well understood. Since lncRNAs are co-expressed with well-annotated coding genes, and gene-gene co-expression strongly correlates with gene function, it is possible to infer the function of all human and mouse lncRNAs using co-expression correlations.[2] Implementing this approach, lncHUB2 is a web-server application that provides known and inferred knowledge about the function of 18,705 human and 11,274 mouse lncRNAs. lncHUB2 produces reports that contain the secondary structure fold of each lncRNA,

☆ This article is part of a special issue entitled: 'Computation Resources (2025)' published in Journal of Molecular Biology.

0022-2836/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

related publications about each lncRNA, the most correlated coding genes, the most correlated lncRNAs, a network that visualizes the most correlated genes, predicted mouse phenotypes, predicted membership in biological processes and pathways, predicted upstream transcription factor regulators, and predicted disease associations. In addition, the reports produced by lncHUB2 include subcellular localization information; expression across tissues, cell types, and cell lines; and predicted small molecules and CRISPR knockouts (CRISPR-KOs) prioritized based on their likelihood to up- or down-regulate the expression of each lncRNA. Most of the knowledge served by the lncHUB2 database is based on gene-gene co-expression correlations computed from publicly available RNA-seq studies from the Gene Expression Omnibus (GEO).[3] Samples from thousands of studies are uniformly processed by the ARCHS4 pipeline[4] to create a gene-gene co-expression correlations matrix. Although lncHUB2 provides new knowledge about lncRNAs, it does not provide the ability to perform enrichment analysis for sets of lncRNAs.

Several lncRNA enrichment analysis tools have been developed.[5–8] For example, the tool for lncRNA set enrichment analysis (TLSEA) uses data from LncRNADisease[9] and lncRNA-miRNA associations to provide background sets for lncRNA-specific human diseases.[6] Similarly, ncPATH is an enrichment analysis tool for ncRNAs that has enrichment analysis functionality against the KEGG database.[8] Hung et al. showed that incorporating lncRNAs in addition to coding genes can improve Gene Ontology and KEGG pathways enrichment analysis.[7] The most comprehensive lncRNA enrichment analysis tool is LncSEA 2.0.[5] The publication and website of LncSEA 2.0 provides support for enrichment analysis against ~400,000 annotated lncRNA gene sets. LncSEA 2.0 gene sets are organized into 86 subcategories, supporting multiple input formats, and providing various visualizations of the enrichment result. Overall, most existing lncRNA enrichment analysis tools suffer from some shortcomings that include low coverage of gene set functions, and crude user interfaces with limited functionality and features. Here we describe lncRNAlyzr, a web-server lncRNA enrichment analysis tool that serves 33 gene set libraries created from Enrichr[10] libraries and gene-gene co-expression correlations from ARCHS4.[4] To visualize the results from the enrichment analysis, lncRNAlyzr uses the Enrichr-KG[11] framework. Finally, to demonstrate lncRNAlyzr, the knockdown of the lncRNA CYTOR in K562 cells followed by RNA-seq profiling is investigated with lncRNAlyzr and Enrichr (Figure 1).

Materials and Methods

Creating the lncRNA set libraries

The lncRNA set libraries were created using 33 annotated protein-coding-genes gene sets libraries from Enrichr.[10] These gene set libraries were transformed into lncRNA set libraries leveraging the gene-gene coexpression data matrix from ARCHS4.[4] The 33 set libraries (Table 1) were selected from the collection of all libraries in Enrichr. These libraries include 26 of the most accessed Enrichr libraries. These 26 libraries were also selected for Enrichr-KG.[11] The other 7 set libraries are newer libraries derived from data collected by Common Fund programs. These libraries were created for the Common Fund Data Ecosystem (CFDE) programs (https://cdfe.info). To convert a gene set into a lncRNA set, we utilized the gene co-expression data matrix from ARCHS4. ARCHS4 has ~2 million uniformly aligned human and mouse samples. From these aligned samples, a subset of 50,000 random samples are selected to compute gene-gene co-expression correlations with the Pearson Correlation Coefficient (PCC) method. From this gene-gene co-expression matrix, we extract a correlation submatrix where the columns are the coding genes from the gene set and the rows are all lncRNAs. We then calculate the mean PCCs for each lncRNA and use this to rank all lncRNAs. For a gene set with n genes, we take the top n most correlated lncRNAs as the new lncRNA set. This process was repeated for every gene set in every library producing new versions of all original gene set libraries with lncRNAs replacing the protein coding genes (Figure 2). Co-expression correlations with the Spearman's Rank Correlation Coefficient method was tested and the results were marginally different.

Knowledge graph database serialization

The lncRNA set libraries are stored in a Neo4j knowledge graph database instance.[12] To ingest the data into the Neo4j database, the lncRNA set libraries were converted into nodes and edges serialization files (Figure 3). The nodes file contains the listing of all lncRNAs and terms that describe lncRNA sets. The uuid5 ID system is implemented to convert the node labels into uuid5 identifiers. Meanwhile, the edges are composed of triples made from the source id, the target id, and the name of a relationship that describe the nature of the association. We define a connection between a lncRNA and a term if the lncRNA belongs to the set of that term. The serialized nodes and edges are then ingested into the Neo4j database using a custom script that calls the Neo4j Python library.

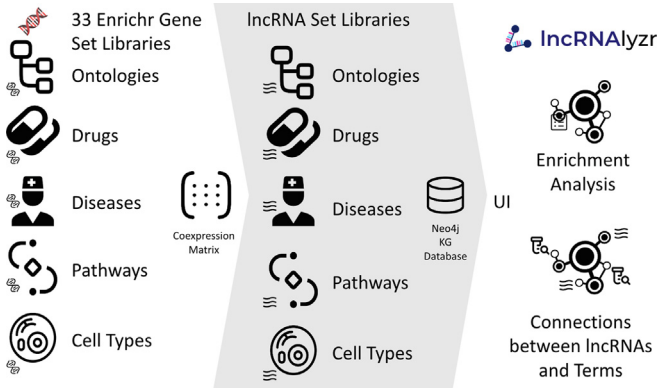


Figure 1. The lncRNAlyzr workflow. lncRNAlyzr provides enrichment analysis for lncRNA sets with 33 gene set libraries that span knowledge about cellular functions, drugs, diseases, pathways, and cell types. To create the lncRNAlyzr database, gene set libraries from Enrichr were converted into lncRNA set libraries and ingested into a Neo4j database. The lncRNAlyzr user interface has functionality for (1) enrichment analysis, (2) querying lncRNAs and their functional terms, and (3) finding paths that connect lncRNAs and their functional terms.

Table 1 lncRNA set libraries stored in the lncRNAlyzr database.

GMT Library	Category	Terms	lncRNA coverage	lncRNAs/term
Achilles_fitness_decrease[60]	Diseases/Drugs	216	4382	128
Achilles_fitness_increase[60]	Diseases/Drugs	216	4510	128
ARCHS4_TFs_Coexp[4]	Transcription	1724	9448	299
CCLE_Proteomics_2020[4,61]	Cell Types	378	5032	586
ChEA_2022[42]	Transcription	757	13,649	1214
Descartes_Cell_Types_and_Tissue_2021[62]	Cell Types	172	4435	139
DisGeNET[63]	Diseases/Drugs	9827	11,114	61
FANTOM6_KD_DEGs[64]	Transcription	350	11,380	342
GlyGen_Glycosylated_Proteins_2022[65]	Pathways	338	1056	46
GO_Biological_Process_2021[43,66]	Ontologies	6036	8312	32
GTEx_Aging_Signatures_2021[57]	Diseases/Drugs	270	2074	250
GTEx_Tissues_V8_2023[57]	Cell Types	511	3600	96
GWAS_Catalog_2019[67]	Diseases/Drugs	1737	6729	36
HuBMAP_ASCTplusB_augmented_2022[68]	Cell Types	777	5462	103
Human_Gene_Atlas[69]	Cell Types	84	6251	258
Human_Phenotype_Ontology[70]	Ontologies	1779	4724	30
IDG_Drug_Targets_2022[71]	Diseases/Drugs	888	1487	15
Jensen_DISEASES[72]	Ontologies	1810	13,441	27
KEGG_2021_Human[46,73]	Pathways	320	3924	101
KOMP2_Mouse_Phenotypes_2022[74]	Ontologies	529	5435	67
LINCS_L1000_Chem_Pert_Consensus_Sigs[75]	Diseases/Drugs	10,850	5209	245
LINCS_L1000_CRISPR_KO_Consensus_Sigs[75]	Diseases/Drugs	10,424	4980	245
Metabolomics_Workbench_Metabolites_2022[76]	Pathways	233	688	14
MGI_Mammalian_Phenotype_Level_4_2021[77]	Ontologies	4601	6840	33
Pfam_InterPro_Domains[78]	Misc	311	2939	35
PFOCR_Pathways[79]	Pathways	17,326	5592	12
Proteomics_Drug_Atlas_2023[80]	Diseases/Drugs	1748	3773	145
Reactome_2022[44]	Pathways	1818	8581	61
Tabula_Muris[81]	Cell Types	106	1848	80
Tabula_Sapiens[82]	Cell Types	469	888	99
The_Kinase_Library_2023[83]	Pathways	303	1814	792
TRRUST_Transcription_Factors_2019[84]	Transcription	568	1986	22
WikiPathway_2021_Human[45]	Pathways	622	3338	43

Developing a dedicated enrichment analysis microservice

lncRNAlyzr uses an independent microservice to perform the enrichment analysis portion of the application. To develop the lncRNA enrichment analysis microservice application, we adapted the code from the Independent Enrichment Analysis Appyter,[13] and converted it into a Flask server that can be accessed via an API. To use the API, we first

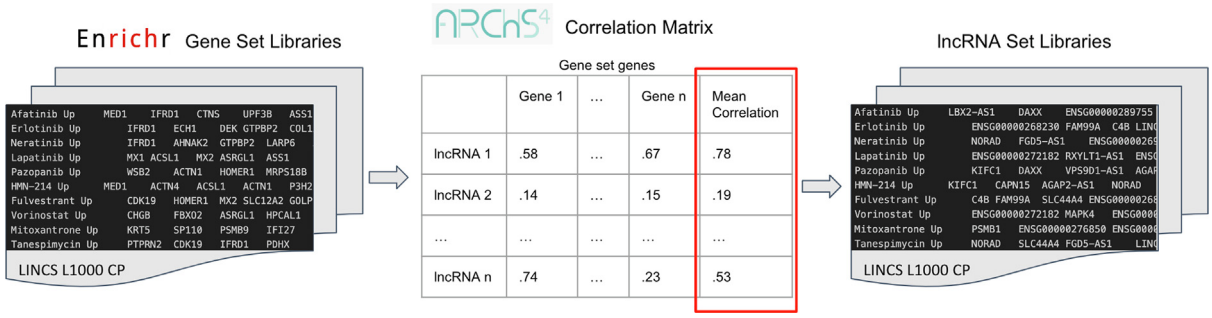


Figure 2. Converting gene set libraries into lncRNA set libraries. 33 gene set libraries from Enrichr were converted into lncRNA set libraries by calculating the correlation between coding genes and lncRNAs using ARCHS4′s gene-gene co-expression correlation data matrix. For each gene in a gene set, a submatrix was extracted where the columns are the members of the gene set and the rows all lncRNAs. The mean correlation coefficient for each row is then computed. The top n lncRNAs ranked by mean correlation coefficient were selected to constitute the matching lncRNA set.

send the lncRNAs to the server to receive a hash identifier. We then send this identifier to the server along with the name of a lncRNA set library to receive enriched lncRNA sets with their *p*-values, overlapping lncRNAs, and other parameters from the enrichment results.

Developing the lncRNAlyzr user interface

The lncRNAlyzr web interface utilizes a custom-built user interface (UI) template we developed for interacting with any network stored in a Neo4j database. The UI component visualizes the results returned from Cypher queries into interactive ball-and-stick diagrams. This user interface component can be customized, and was employed for other projects including Reprotox-KG[14] and Enrichr-KG.[11] The user interface is built using the Next.js, React, and Material UI framework. We also leveraged Cytoscape.js[15] to produce the visualizations of the subnetworks returned by the Cypher queries.

Cell culture

K562 leukemia cells were maintained in Roswell Park Memorial Institute (RPMI) 1640 medium (Gibco), supplemented with 10% fetal calf serum (FCS), 1% L-Glutamine, and 1% pen strep antibiotics (Biological Industries). Cells were incubated at 37 °C with 5% CO2. Cell count and viability were measured through trypan blue staining by the Luna Automated Cell Counter. Mycoplasma contamination was tested regularly.

sgRNA design

The sgRNAs were designed using the CHOPCHOP website (https://chopchop.cbu.uib.no/#).[16,17] For targeting the lncRNA CYTOR, two specific sgRNAs were selected: sgRNA1 (GAGGGAAATAAATGACTGGA) and sgRNA3 (GTGTACATCATTGGGAATGG). Additionally, a sgRNA, NC10010 (GTCCACCCTTATCTAGGCTA), was used as a Negative Control (NC); this sgRNA was non-targeting to the human genome.

Cloning sgRNA sequences into plasmids

The protospacer sequence were cloned into the following sgRNA backbone plasmids: pSB700 was a gift from George Church (Addgene plasmid #64046; https://n2t.net/addgene:64046; RRID: Addgene 64046),[18] with different options of selection (puromycin resistance, blasticidin resistance). Approximately 100 ng of the plasmid were digested by the appropriate restriction enzyme (Esp3I nuclease (NEB) for psb700 plasmids, BPU11021 and BstXI for pCRISPRia-v2) and sgRNA oligonucleotides were annealed and ligated into the digested plasmid by T4 DNA Ligase (NEB). Ligated plasmids were then transformed into E. coli DH5α competent cells through heat-shock, which were then selected by ampicillin. Colonies were picked, and the ligated plasmid was extracted using the Plasmid DNA Miniprep Kit (Bio Basic) according to the manufacturer instructions. The extracted plasmid was sequenced by Macrogen to confirm presence of the sgRNA oligonucleotide.

Polyjet transfection and polybrene transduction of CRISPRi system and sgRNAs

Lentiviral transfection was performed using Polyjet In Vitro Transfection Reagent (SignaGen, Frederick, MD, USA) following the manufacturer protocol. Briefly, Hek293 cells were seeded 24 hrs. before transfection at a concentration to obtain a confluency of 90% at time of transfection. The cells were transfected with three plasmids: the plasmid of interest, and two lentiviral packaging plasmids (VSVG (Addgene plasmid #8454) and psPAX2 (Addgene plasmid #12260)). Lentiviral transduction was performed 48 hrs. later using Polybrene Infection Reagent (Sigma) by

harvesting the synthesized virus and infecting the k562 cell line stably expressing the dCas9-KRAB system. As positive control of both transfection and transduction, a plasmid containing GFP was used. GFP expression was confirmed through fluorescence microscopy. Stable transduced cells were selected with blasticidin, puromycin. Only one sgRNA was used per reaction. The k562 cell line stably expressing the dCas9-KRAB system was produced by transfecting Hek293 cells with two lentiviral packaging plasmids (VSVG and psPAX), and the pHR-SFFV-dCas9-BFP-KRAB CRISPRi cassette (pHR-SFFV-dCas9-BFP-KRAB was a gift from Stanley Qi & Jonathan Weissman (Addgene plasmid #46911; https://n2t.net/addgene:46911; RRID: Addgene_46911).[18,19] The synthesized viruses were then used to transduce k562 cells. Positive cells were selected according to BFP fluorescence using the BD FACS Aria-IIIU Sorter.

3′ CEL-seq

The samples containing sgRNAs that silenced lncRNA CYTOR, along with the NC, were produced in four replicates and analyzed through CEL-seq. The transcriptomes of these selected samples were quantified by 3′ CEL-seq by the Technion Genomics Center. Samples were barcoded and pooled before linear amplification of mRNA. During sample preparation, each mRNA transcript was tagged with a Unique Molecular Identifier (UMI). RNA was extracted with the PureLink RNA Mini Kit (Thermo Fisher Scientific). The quality of the extracted RNA was validated by 1.5% agarose gel (120 V for 45 min). RNA samples were quantified using Qubit Flex (Invitrogen) using the RNA HS assay kit (cat no. Q32855). The quality of the RNA was evaluated using the TapeStation 4200 (Agilent) with the RNA kit (cat no. 5067—5576). The RINe values of all samples were in the range of 9.5–10.0, indicating high quality. CEL-Seq2 sequencing libraries were prepared as published by Hashimshony et al.,[20] with one modification- instead of single cells as input, 2 ng purified RNA was taken as input for library preparation. The CEL-Seq2 data was generated on Illumina NextSeq2000, using P2 100 cycles (Read1-12; Index1-6; Index2-0; Read2-65) (Illumina, cat no. 20046811). Demultiplexing of the data was performed in two steps. First, Illumina demultiplexing was performed using bcl2fastq Illumina software with the following parameters: barcode-mismatches = 1, minimum-trimmed-read-length = 0, and mask-short-adapter-reads = 0. Second, CEL-Seq2 demultiplexing, using the pipeline described in Hashimshony et al. (https://github.com/yanailab/CEL-Seq-pipeline), was executed with the following parameters: min_bc_quality = 10, bc_length = 6, umi_length = 6, and cut_length = 65. Raw data- FASTQ files for each sample were received. Sequencing reads were aligned to the human genome reference GRCh38 primary assembly. After undergoing quality assurance tests, the sequences obtained from the CEL-seq were aligned and matched to their appropriate gene. The number of reads per gene was recorded for each of the samples.

Results

Exploring lncRNA set functions with lncRNAlyzr

lncRNAlyzr is a web-server application that enables users to query a set of lncRNAs to find possible common functions among them. Since knowledge about lncRNAs is limited, lncRNAlyzr utilizes gene-gene co-expression correlations to systematically predict lncRNA functions based on

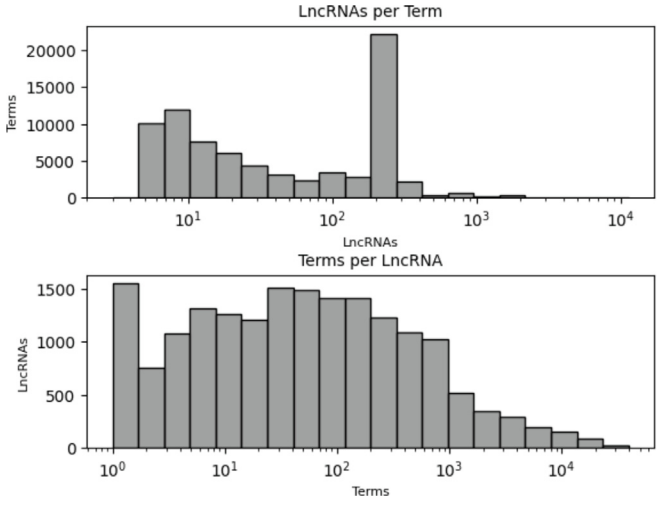


Figure 3. LncRNAlyzr database node and edge distributions. (A) lncRNA set size per term. (B) Terms per lncRNA.

their co-expression with human coding genes. To create the lncRNA set libraries, we leverage 33 annotated gene sets from Enrichr.[10] These gene set libraries are converted into lncRNA set libraries using RNA-seq gene-gene co-expression correlations obtained from ARCHS4.[4] lncRNAlyzr was developed by adapting the user interface previously used to create Reprotox-KG,[14] Enrichr-KG,[11] and Harmonizome-KG.[21] The KG-UI provides the following functions: (1) integrated enrichment analysis for lncRNA sets; (2) single node search for querying the immediate connections between lncRNAs and their functional terms; and (3) two node searches for querying the shortest paths between terms, lncRNAs, or both. All these features are made possible by serializing the lncRNA set libraries into nodes and edges and ingesting them into a Neo4j knowledge graph database. Examining the distributions of lncRNAs and their functional terms (Figure 3), we observed a peak of gene sets of size 250, this is because of an arbitrary cutoff applied to some of the Enrichr gene sets. We also see that many lncRNAs are present in few gene sets.

To perform enrichment analysis, users can upload their own lncRNA sets into an input text box. The user-inputted lncRNAs are automatically validated against the lncRNAs in the lncRNAlyzr database. Users then need to select up to five lncRNA set libraries for performing the enrichment analysis. Alternatively, users can click on the resource icons to select libraries created from each resource (Figure S1). Upon pressing the submit button, lncRNAlyzr returns a subnetwork made of the enriched terms from the selected

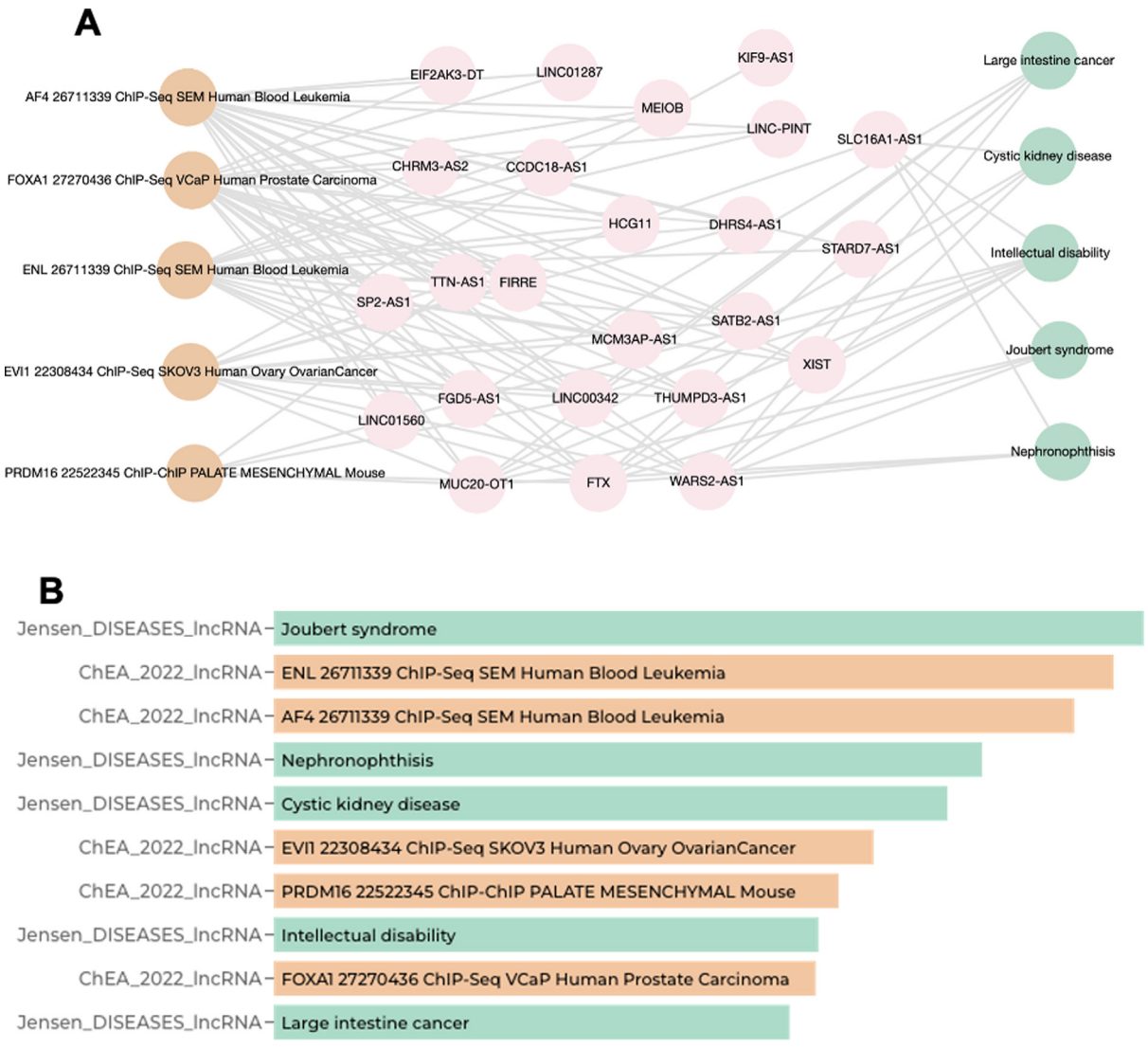


Figure 4. Enrichment analysis applied to the up-regulated lncRNAs after CYTOR knockdown. (A) Subnetworks that connect enriched terms and their overlapping up regulated lncRNAs. (B) The same results are visualized as a bar chart.

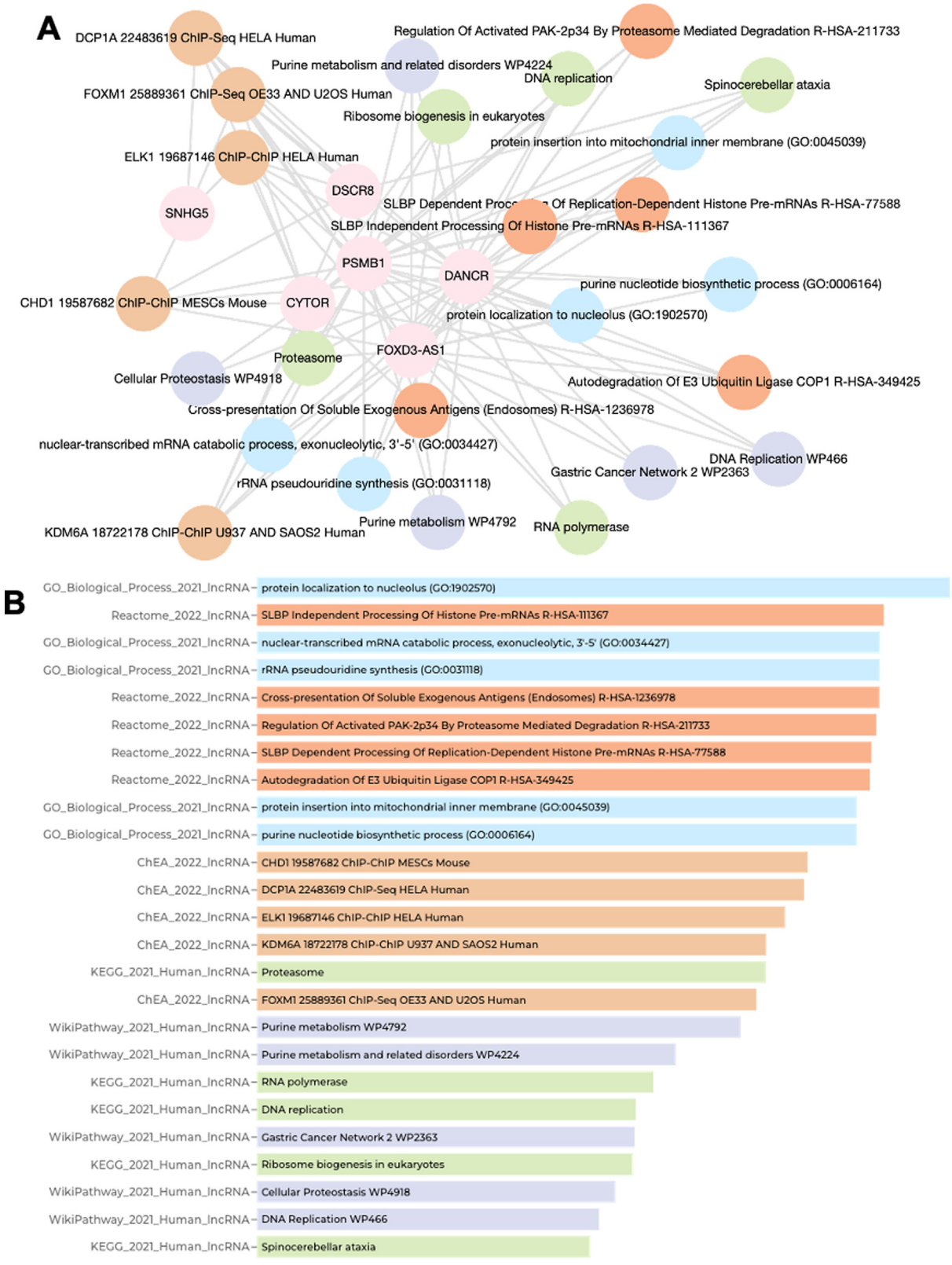


Figure 5. Enrichment analysis applied to the up-regulated coding genes after CYTOR knockdown. (A) Subnetworks that connect enriched terms and their overlapping up regulated coding genes. (B) The same results are visualized as a bar chart.

libraries connected to the lncRNAs that overlap with these enriched terms (Figure S2). The subnetwork can be modified by adding or removing libraries, adjusting the size limit, as well as by changing other parameters such as maximum node degree. The layout of the subnetwork can be modified by choosing one of three available layouts: force-directed, geometrical, or hierarchical. The subnetwork can be downloaded as an image file, or as a serialized nodes and edges files containing additional metadata. The top enriched terms can also be viewed as a table or a bar chart.

Radiation sensitivity after CYTOR knockdown use case

Radiation therapy is one of the most common treatments for cancer therapy. Typically, such treatment is combined with chemotherapy. Overall, this approach has been successful for short term cancer management. However, in many cases tumors become radio- and chemo-resistant, dampening the benefits of radiation and chemotherapy. Hence, it is imperative to discover the mechanisms involved in radio- and chemo-resistance. LncRNAs are known to play an



Figure 6. Enrichment analysis applied to the down-regulated lncRNAs after CYTOR knockdown. (A) Subnetworks that connect enriched terms and their overlapping down regulated lncRNAs. (B) The same results are visualized as a bar chart.

important role in radioresistance, chemoresistance, and tumor progression. One of the most studied among these lncRNA regulators is the cytoskeleton regulator RNA (CYTOR).[22] CYTOR has been linked to metastasis,[23–29] energy metabolism, glycolysis,[30] chemoresistance,[31,32] radioresistance,[33] cell proliferation,[26,34–36] and inflammatory responses.[22,37] Dysregulation of CYTOR was linked to poor prognosis in several cancers.[22,38] Hence, to gain a better understanding about the functional roles of CYTOR, we analyzed the effects of knocking down CYTOR with CRISPRi in the K562 leukemia cell line.

CRISPR interference targeting of CYTOR was performed in the K562 cell line. Bulk RNA-seq was then applied to profile the global changes in gene expression due to the knockdown. Differential expression analysis was performed using Biojupies[39]. BioMart[40] was used to separate protein-coding genes from long non-coding RNAs. The differentially expressed genes and lncRNAs were divided into up- and down-regulated protein coding gene sets (limma voom,[41] p-value < 0.05 and |log2 fold change| > 0) and differentially expressed lncRNA sets (p-value < 0.05 and |log2 fold change| > 0). The up and down regulated lncRNAs are then analyzed by lncRNAlyzr with the lncRNA libraries (1) ChEA for transcription factors enrichment analysis,[42] (2) Gene Ontology (GO) biological processes,[43] (3) Reactome,[44] (4) WikiPathways,[45] and (5) KEGG[46] for pathway enrichment analysis (Figures 4 and 5). Additionally, the up and down protein coding gene sets were analyzed with Enrichr-KG[11] using the same libraries (Figures 6 and 7).

Enrichment analysis of the up-regulated lncRNAs (Figure 4) and coding genes (Figure 6) after CYTOR knockdown identified high overlap with target genes of the transcription factors AF4 and ENL (p < 1.0e−7, Fisher's exact test) from a ChIP-seq study that profiled these transcription factors in the RS4;11 lymphoblast cell line.[47] AF4 and ENL are members of the P-TEFb complex[48–50] that regulates transcription elongation, pre-mRNA processing, and histone modifications.[51,52] Hence, it is possible that CYTOR activity dampens this complex which is known to promote cancer progression. Other themes that emerge from both enrichment analyses of the up-regulated coding and non-coding gene sets is the relevance of CYTOR to cancer, and specifically cancer of the kidney. The gene set library created from the DISEASES database ranked many cancers as the most enriched terms (Figure 6) by Enrichr-KG. At the same time, lncRNAlyzr ranked nephronophthisis and cystic kidney disease (p < 1.0e−6, Fisher's exact test) as two of the most enriched terms (Figure 4). Clearly, these insights are just two examples out of the many enriched terms that were identified. It should be noted that the consistency between the coding and non-coding terms suggest that the identified differentially up regulated lncRNAs are indeed commonly co-expressed with the differentially expressed coding genes. In contrast, enrichment analysis of the down-regulated lncRNAs (Figure 6) and coding genes (Figure 7) after CYTOR knockdown shows weaker agreement. The enrichment analysis of the down-regulated coding gene displays clear signal for down-regulation of mitochondrial related functions including fatty acid beta oxidation from Reactome (R-HSA-77289, Fisher's exact test, p < 0.0008) and mitochondrial fission from the Gene Ontology (GO:0000266, p < 0.003). The transcription factor Myc is also enriched for the down-regulated genes based on the ChEA[42] library (Myc ChIP-seq studies,[53,54] p < 1.0e−10). The only top enriched term in lncRNAlyzr that agrees with Enrichr-KG is protein insertion into the mitochondrial inner membrane from the Gene Ontology (GO:0045039, p < 1.0e−6). Overall, it appears that the terms identified by lncRNAlyzr are more specific. The less clear picture for the down-regulated coding and non-coding genes after CYTOR knockdown might also be due to the observation that the CYTOR knockdown induces up regulation of many more genes compared to the down regulation of genes. This may suggest that CYTOR is acting more as a transcriptional repressor than promoter.

Discussion and Conclusions

Here we present lncRNAlyzr, a web application that provides enrichment analysis for lncRNA sets across a wide-array of libraries created by combining gene set libraries from coding genes with gene-gene co-expression correlations made from publicly available bulk RNA-seq studies. This approach has several limitations. First, the application ignores information that is available about lncRNAs in the literature. Such information could be added to the application as additional assertions added to the KG database. Second, the gene-gene correlations matrix used for the predictions is made from studies conducted in many different tissues and cell lines, masking the cell and tissue specific roles of genes and lncRNAs. We previously addressed this issue by partitioning the ARCHS4 data into clusters and then making cluster specific gene-gene co-expression matrices.[55] We can also create context specific gene-gene co-expression correlations from other resources such as HuBMAP,[56] GTEx,[57] CPTAC[58] and TCGA.[59] We intend to apply this approach to lncRNA sets in the future. Another limitation is that the functional predictions use the PCC method which makes such predictions using a linear unsupervised learning method. More sophisticated nonlinear machine learning methods to create the lncRNA libraries should be possible. Another limitation of lncRNAlyzr is that it provides enrichment analysis for lncRNAs separately from coding genes. LncRNAs function together with cod-

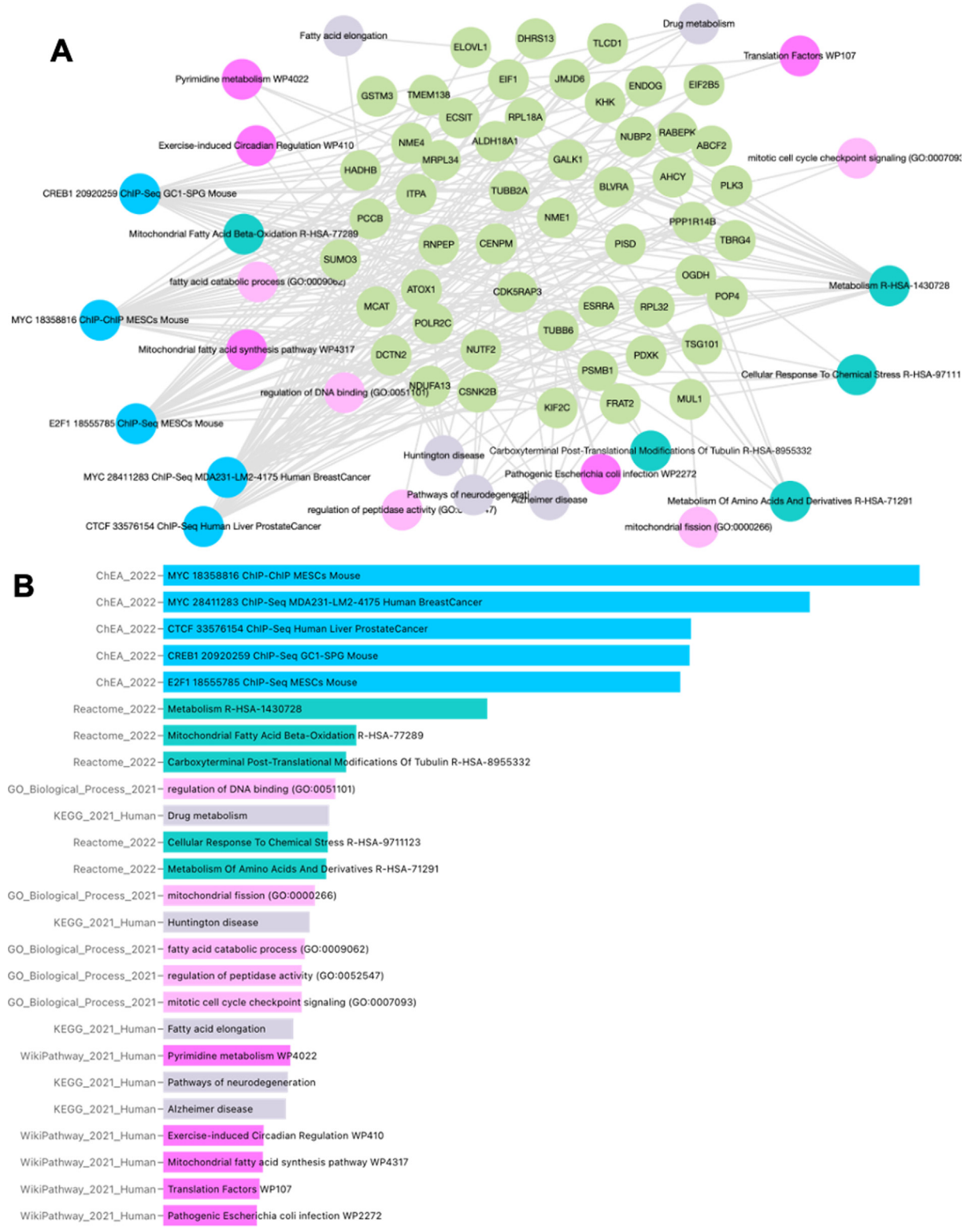


Figure 7. Enrichment analysis applied to the down-regulated coding genes after CYTOR knockdown. (A) Subnetworks that connect enriched terms and their overlapping down regulated coding genes. (B) The same results are visualized as a bar chart.

ing genes to carry out their cellular roles. Hence, an integrative analysis that combines lncRNAs together with coding genes might be more insightful than looking at lncRNAs alone. This limitation of the application may be relevant to some use cases, while for others, keeping the analysis separate

might be more beneficial for the investigation. For this reason, as shown in the CYTOR KD use case, it is recommended to combine analysis with lncRNAlyzr with Enrichr analysis to obtain a more complete picture when analyzing differentially expressed genes. In other cases, for example, for analyzing data from lncRNA knockout screens, lncRNAlyzr should be a useful tool to understand the collective function of a set of positive lncRNAs hits.

Data and code availability

The lncRNAylzr web-server application is available at: https://lncrnalyzr.maayanlab.cloud/.

The lncRNA libraries in GMT format are available at: https://lncrnalyzr.maayanlab.cloud/libraries.

The lncRNAylzr source code is available from: https://github.com/MaayanLab/lncRNAlyzr.

The CYTOR KO RNA-seq dataset is available from the Gene Expression Omnibus.

(Accession # GSE285084).

Funding sources

The project is partially funded by the NIH grants R01DK131525, OT2OD036435, OT2OD030160, U24CA264250, U24CA271114, and RC2DK131995 to AM; and Israeli Science Foundation (ISF; Grant No. 2228/19) and the Israel Cancer Association (ICA; Grant No. 20230069 and 2020095) to AB.

CRediT authorship contribution statement

John Erol Evangelista: Writing – original draft, Visualization, Software, Formal analysis, Data curation. **Tahleel Ali-Nasser:** Writing – original draft, Validation, Resources, Methodology, Investigation. **Lauren E. Malek:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Zhuorui Xie:** Software, Methodology, Investigation, Formal analysis. **Giacomo B. Marino:** Software, Resources, Investigation, Formal analysis. **Assaf C. Bester:** Writing – original draft, Validation, Resources, Methodology, Funding acquisition, Formal analysis. **Avi Ma'ayan:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization.

DATA AVAILABILITY

The CYTOR KO RNA-seq dataset is available from the Gene Expression Omnibus (Accession # GSE285084).

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary material to this article can be found online at https://doi.org/10.1016/j.jmb.2025.168938.

Received 14 November 2024;
Accepted 6 January 2025;
Available online 11 January 2025

Keywords:
lncRNA;
enrichment analysis;
knowledge graphs;
Enrichr-KG;
Enrichr

References

1. Schmitt, A.M., Chang, H.Y., (2016). Long noncoding RNAs in cancer pathways. *Cancer Cell* **29**, 452–463.
2. Marino, G.B., Wojciechowicz, M.L., Clarke, D.J.B., Kuleshov, M.V., Xie, Z., Jeon, M., Lachmann, A., Ma'ayan, A., (2023). lncHUB2: aggregated and inferred knowledge about human and mouse lncRNAs. *Database* **2023** https://doi.org/10.1093/database/baad009.
3. Clough, E., Barrett, T., (2016). The gene expression omnibus database. *Methods Mol. Biol.* **1418**, 93–110.
4. Lachmann, A., Torre, D., Keenan, A.B., Jagodnik, K.M., Lee, H.J., Wang, L., Silverstein, M.C., Ma'ayan, A., (2018). Massive mining of publicly available RNA-seq data from human and mouse. *Nature Commun.* **9**, 1366.
5. Zhang, G., Song, C., Fan, S., Yin, M., Wang, X., Zhang, Y., Huang, X., Li, Y., Shang, D., Li, C., Wang, Q., (2024). LncSEA 2.0: an updated platform for long non-coding RNA related sets and enrichment analysis. *Nucleic Acids Res.* **52**, D919–D928.
6. Li, J., Li, Z., Wang, Y., Lin, H., Wu, B., (2023). TLSEA: a tool for lncRNA set enrichment analysis based on multi-source heterogeneous information fusion. *Front. Genet.* **14**, 1181391
7. Hung, K.-S., Hsiao, C.-C., Pai, T.-W., Hu, C.-H., Tzou, W.-S., Wang, W.-D., Chen, Y.-R., (2018). Functional enrichment analysis based on long noncoding RNA associations. *BMC Syst. Biol.* **12**, 45.

8. Li, Z., Zhang, Y., Fang, J., Xu, Z., Zhang, H., Mao, M., Chen, Y., Zhang, L., Pian, C., (2023). NcPath: a novel platform for visualization and enrichment analysis of human non-coding RNA and KEGG signaling pathways. *Bioinformatics* **39** https://doi.org/10.1093/bioinformatics/btac812.
9. Lin, X., Lu, Y., Zhang, C., Cui, Q., Tang, Y.-D., Ji, X., Cui, C., (2024). LncRNADisease v3.0: an updated database of long non-coding RNA-associated diseases. *Nucleic Acids Res.* **52**, D1365–D1369.
10. Chen, E.Y., Tan, C.M., Kou, Y., Duan, Q., Wang, Z., Meirelles, G.V., Clark, N.R., Ma'ayan, A., (2013). Enrichr: interactive and collaborative HTML5 gene list enrichment analysis tool. *BMC Bioinf.* **14**, 128.
11. Evangelista, J.E., Xie, Z., Marino, G.B., Nguyen, N., Clarke, D.J.B., Ma'ayan, A., (2023). Enrichr-KG: bridging enrichment analysis across multiple libraries. *Nucleic Acids Res.* **51**, W168–W179.
12. Lyon, W., (2022). Full stack GraphQL applications: with react, Node.js, and Neo4j. Simon and Schuster.
13. Clarke, D.J.B., Jeon, M., Stein, D.J., Moiseyev, N., Kropiwnicki, E., Dai, C., Xie, Z., Wojciechowicz, M.L., Litz, S., Hom, J., Evangelista, J.E., Goldman, L., Zhang, S., Yoon, C., Ahamed, T., Bhuiyan, S., Cheng, M., Karam, J., Jagodnik, K.M., Shu, I., Lachmann, A., Ayling, S., Jenkins, S.L., Ma'ayan, A., (2021). Appyters: turning jupyter notebooks into data-driven web apps. *Patterns (N Y)* **2**, 100213
14. Evangelista, J.E., Clarke, D.J.B., Xie, Z., Marino, G.B., Utti, V., Jenkins, S.L., Ahooyi, T.M., Bologa, C.G., Yang, J.J., Binder, J.L., Kumar, P., Lambert, C.G., Grethe, J.S., Wenger, E., Taylor, D., Oprea, T.I., de Bono, B., Ma'ayan, A., (2023). Toxicology knowledge graph for structural birth defects. *Commun Med (Lond)* **3**, 98.
15. Franz, M., Lopes, C.T., Huck, G., Dong, Y., Sumer, O., Bader, G.D., (2015). Cytoscape.js: a graph theory library for visualisation and analysis. *Bioinformatics* **32**, 309–311.
16. Montague, T.G., Cruz, J.M., Gagnon, J.A., Church, G.M., Valen, E., (2014). CHOPCHOP: a CRISPR/Cas9 and TALEN web tool for genome editing. *Nucleic Acids Res.* **42**, W401–W407.
17. Labun, K., Montague, T.G., Krause, M., Torres Cleuren, Y. N., Tjeldnes, H., Valen, E., (2019). CHOPCHOP v3: expanding the CRISPR web toolbox beyond genome editing. *Nucleic Acids Res.* **47**, W171–W174.
18. Chavez, A., Scheiman, J., Vora, S., Pruitt, B.W., Tuttle, M., Iyer, E.P.R., Lin, S., Kiani, S., Guzman, C.D., Wiegand, D. J., Ter-Ovanesyan, D., Braff, J.L., Davidsohn, N., Housden, B.E., Perrimon, N., Weiss, R., Aach, J., Collins, J.J., Church, G.M., (2015). Highly efficient Cas9-mediated transcriptional programming. *Nature Methods* **12**, 326–328.
19. Gilbert, L.A., Larson, M.H., Morsut, L., Liu, Z., Brar, G.A., Torres, S.E., Stern-Ginossar, N., Brandman, O., Whitehead, E.H., Doudna, J.A., Lim, W.A., Weissman, J. S., Qi, L.S., (2013). CRISPR-mediated modular RNA-guided regulation of transcription in eukaryotes. *Cell* **154**, 442–451.
20. Hashimshony, T., Senderovich, N., Avital, G., Klochendler, A., de Leeuw, Y., Anavy, L., Gennert, D., Li, S., Livak, K.J., Rozenblatt-Rosen, O., Dor, Y., Regev, A., Yanai, I., (2016). CEL-Seq2: sensitive highly-multiplexed single-cell RNA-Seq. *Genome Biol.* **17**, 77.
21. Diamant, I., Clarke, D.J.B., Evangelista, J.E., Lingam, N., Ma'ayan, A., (2024). Harmonizome 3.0: integrated knowledge about genes and proteins from diverse multi-omics resources. *Nucleic Acids Res.*. https://doi.org/10.1093/nar/gkae1080.
22. Ou, C., He, X., Liu, Y., Zhang, X., (2023). lncRNA cytoskeleton regulator RNA (CYTOR): Diverse functions in metabolism, inflammation and tumorigenesis, and potential applications in precision oncology. *Genes Dis* **10**, 415–429.
23. Zhu, Z., Lu, J., Tong, J., Yin, Y., Zhang, K., (2024). LncRNA CYTOR knockdown inhibits tumor development via regulating miR-503-5p/PCSK9 in lung adenocarcinoma. *Am. J. Med. Sci.* **368**, 382–391.
24. Wang, W., Yun, B., Hoyle, R.G., Ma, Z., Zaman, S.U., Xiong, G., Yi, C., Xie, N., Zhang, M., Liu, X., Bandyopadhyay, D., Li, J., Wang, C., (2024). CYTOR facilitates formation of FOSL1 phase separation and super enhancers to drive metastasis of tumor budding cells in head and neck squamous cell carcinoma. *Adv. Sci. (Weinh.)* **11**, e2305002
25. Li, J., Sun, K., Zhang, M., Luo, Y., Zhang, Y., Huang, C., Hong, J., Chen, D., Liao, W., (2023). Long non-coding RNA enhances gastric carcinoma proliferation, migration and invasion via the miR-136-5p/HOXC10 axis. *Am. J. Cancer Res.* **13**, 2714–2731.
26. Keqi, H., Handong, L., (2021). The Long Non-coding RNA Cytoskeleton Regulator (CYTOR) Sponges microRNA-206 (miR-206) to Promote Proliferation and Invasion of HP75 Cells. *Curr. Cancer Drug Targets* **21**, 526–535.
27. Wu, T., Zhang, D.-L., Wang, J.-M., Jiang, J.-Y., Du, X., Zeng, X.-Y., Du, Z.-X., (2020). TRIM29 inhibits miR-873-5P biogenesis via CYTOR to upregulate fibronectin 1 and promotes invasion of papillary thyroid cancer cells. *Cell Death Dis.*. **11**, 813.
28. Reon, B.J., Takao Real Karia, B., Kiran, M., Dutta, A., (2018). Promotes invasion through a 3'-hairpin structure and associates with prognosis in glioblastoma. *Mol. Cancer Res.* **16**, 1470–1482.
29. Yue, B., Liu, C., Sun, H., Liu, M., Song, C., Cui, R., Qiu, S., Zhong, M., (2018). A positive feed-forward loop between LncRNA-CYTOR and Wnt/β-catenin signaling promotes metastasis of colon cancer. *Mol. Ther.* **26**, 1287.
30. Zhu, W., Wang, J., Liu, X., Xu, Y., Zhai, R., Zhang, J., Wang, M., Wang, M., Liu, L., (2022). lncRNA CYTOR promotes aberrant glycolysis and mitochondrial respiration via HNRNPC-mediated ZEB1 stabilization in oral squamous cell carcinoma. *Cell Death Dis.* **13**, 703.
31. Cao, Q., Wang, H., Zhu, J., Qi, C., Huang, H., Chu, X., (2024). lncRNA CYTOR promotes lung adenocarcinoma gemcitabine resistance and epithelial-mesenchymal transition by sponging miR-125a-5p and upregulating ANLN and RRM2. *Acta Biochim. Biophys. Sin. (Shanghai)*. https://doi.org/10.3724/abbs.2024113.
32. Liu, Y., Li, M., Yu, H., Piao, H., (2020). lncRNA CYTOR promotes tamoxifen resistance in breast cancer cells via sponging miR 125a 5p. *Int. J. Mol. Med.* **45**, 497–509.
33. Jiang, G., Yu, H., Li, Z., Zhang, F., (2021). lncRNA cytoskeleton regulator reduces non small cell lung cancer radiosensitivity by downregulating miRNA 206 and activating prothymosin α,. *Int. J. Oncol.* **59** https://doi.org/10.3892/ijo.2021.5268.

34. Wu, Z.-Y., Wang, Y., Hu, H., Ai, X.-N., Zhang, Q., Qin, Y.-G., (2023). Long noncoding RNA cytoskeleton regulator RNA suppresses apoptosis in hepatoma cells by modulating the miR-125a-5p/HS1-associated protein X-1 axis to induce caspase-9 inactivation. *Gut Liver* **17**, 916–925.
35. Xu, X., Chen, Y., Zhang, Y., Cai, H., Shen, P., Peng, J., Liu, H., Chen, X., Chu, F., (2023). CYTOR promotes proliferation of lung cancer cell by targeting miR-103a-3p to upregulate HMGB1. *Mol. Biotechnol.* **65**, 1528–1538.
36. Tian, Q., Yan, X., Yang, L., Liu, Z., Yuan, Z., Zhang, Y., (2021). lncRNA CYTOR promotes cell proliferation and tumor growth via miR-125b/SEMA4C axis in hepatocellular carcinoma. *Oncol. Lett.* **22**, 796.
37. Lim, J.J., Vining, K.H., Mooney, D.J., Blencowe, B.J., (2024). Matrix stiffness-dependent regulation of immunomodulatory genes in human MSCs is associated with the lncRNA CYTOR. *PNAS* **121**, e2404146121
38. Hernández, L., Nadeu, F., Delgado, J., Duran-Ferrer, M., Romero, D.R., Fernández, R.O., Bittolo, T., Bomben, R., Gattei, V., Campo, E., (2023). P608: the expression of cytor lncrna has poor prognostic value in CLL patients and is associated with microenvironmental stimuli. *HemaSphere* **7**, e4399183
39. Torre, D., Lachmann, A., Ma'ayan, A., (2018). BioJupies: Automated generation of interactive notebooks for RNA-seq data analysis in the cloud. *Cell Syst.* **7**, 556–561.e3.
40. Kinsella, R.J., Kähäri, A., Haider, S., Zamora, J., Proctor, G., Spudich, G., Almeida-King, J., Staines, D., Derwent, P., Kerhornou, A., Kersey, P., Flicek, P., (2011). Ensembl BioMarts: a hub for data retrieval across taxonomic space. *Database (Oxford)* **2011**, bar030
41. Law, C.W., Chen, Y., Shi, W., Smyth, G.K., (2014). voom: precision weights unlock linear model analysis tools for RNA-seq read counts. *Genome Biol.* **15**, R29.
42. Keenan, A.B., Torre, D., Lachmann, A., Leong, A.K., Wojciechowicz, M.L., Utti, V., Jagodnik, K.M., Kropiwnicki, E., Wang, Z., Ma'ayan, A., (2019). ChEA3: transcription factor enrichment analysis by orthogonal omics integration. *Nucleic Acids Res.* **47**, W212–W224.
43. Gene Ontology Consortium, Aleksander, S.A., Balhoff, J., Carbon, S., Cherry, J.M., Drabkin, H.J., Ebert, D., Feuermann, M., Gaudet, P., Harris, N.L., Hill, D.P., Lee, R., Mi, H., Moxon, S., Mungall, C.J., Muruganugan, A., Mushayahama, T., Sternberg, P.W., Thomas, P.D., Van Auken, K., Ramsey, J., Siegele, D.A., Chisholm, R.L., Fey, P., Aspromonte, M.C., Nugnes, M.V., Quaglia, F., Tosatto, S., Giglio, M., Nadendla, S., Antonazzo, G., Attrill, H., Dos Santos, G., Marygold, S., Strelets, V., Tabone, C.J., Thurmond, J., Zhou, P., Ahmed, S.H., Asanitthong, P., Luna Buitrago, D., Erdol, M.N., Gage, M.C., Ali Kadhum, M., Li, K.Y.C., Long, M., Michalak, A., Pesala, A., Pritazahra, A., Saverimuttu, S.C.C., Su, R., Thurlow, K. E., Lovering, R.C., Logie, C., Oliferenko, S., Blake, J., Christie, K., Corbani, L., Dolan, M.E., Drabkin, H.J., Hill, D. P., Ni, L., Sitnikov, D., Smith, C., Cuzick, A., Seager, J., Cooper, L., Elser, J., Jaiswal, P., Gupta, P., Jaiswal, P., Naithani, S., Lera-Ramirez, M., Rutherford, K., Wood, V., De Pons, J.L., Dwinell, M.R., Hayman, G.T., Kaldunski, M. L., Kwitek, A.E., Laulederkind, S.J.F., Tutaj, M.A., Vedi, M., Wang, S.-J., D'Eustachio, P., Aimo, L., Axelsen, K., Bridge, A., Hyka-Nouspikel, N., Morgat, A., Aleksander, S.A., Cherry, J.M., Engel, S.R., Karra, K., Miyasato, S.R., Nash, R.S., Skrzypek, M.S., Weng, S., Wong, E.D., Bakker, E., Berardini, T.Z., Reiser, L., Auchincloss, A., Axelsen, K., Argoud-Puy, G., Blatter, M.-C., Boutet, E., Breuza, L., Bridge, A., Casals-Casas, C., Coudert, E., Estreicher, A., Livia Famiglietti, M., Feuermann, M., Gos, A., Gruaz-Gumowski, N., Hulo, C., Hyka-Nouspikel, N., Jungo, F., Le Mercier, P., Lieberherr, D., Masson, P., Morgat, A., Pedruzzi, I., Pourcel, L., Poux, S., Rivoire, C., Sundaram, S., Bateman, A., Bowler-Barnett, E., Bye-A-Jee, H., Denny, P., Ignatchenko, A., Ishtiaq, R., Lock, A., Lussi, Y., Magrane, M., Martin, M.J., Orchard, S., Raposo, P., Speretta, E., Tyagi, N., Warner, K., Zaru, R., Diehl, A. D., Lee, R., Chan, J., Diamantakis, S., Raciti, D., Zarowiecki, M., Fisher, M., James-Zorn, C., Ponferrada, V., Zorn, A., Ramachandran, S., Ruzicka, L., Westerfield, M., (2023). The gene ontology knowledgebase in 2023. *Genetics* **224** https://doi.org/10.1093/genetics/iyad031.
44. Milacic, M., Beavers, D., Conley, P., Gong, C., Gillespie, M., Griss, J., Haw, R., Jassal, B., Matthews, L., May, B., Petryszak, R., Ragueneau, E., Rothfels, K., Sevilla, C., Shamovsky, V., Stephan, R., Tiwari, K., Varusai, T., Weiser, J., Wright, A., Wu, G., Stein, L., Hermjakob, H., D'Eustachio, P., (2023). The reactome pathway knowledgebase 2024. *Nucleic Acids Res.* **52**, D672–D678.
45. Agrawal, A., Balcı, H., Hanspers, K., Coort, S.L., Martens, M., Slenter, D.N., Ehrhart, F., Digles, D., Waagmeester, A., Wassink, I., Abbassi-Daloii, T., Lopes, E.N., Iyer, A., Acosta, J.M., Willighagen, L.G., Nishida, K., Riutta, A., Basaric, H., Evelo, C.T., Willighagen, E.L., Kutmon, M., Pico, A.R., (2024). WikiPathways 2024: next generation pathway database. *Nucleic Acids Res.* **52**, D679–D689.
46. Kanehisa, M., Goto, S., (2000). KEGG: Kyoto encyclopedia of genes and genomes. *Nucleic Acids Res.* **28**, 27–30.
47. Benito, J.M., Godfrey, L., Kojima, K., Hogdal, L., Wunderlich, M., Geng, H., Marzo, I., Harutyunyan, K.G., Golfman, L., North, P., Kerry, J., Ballabio, E., Chonghaile, T.N., Gonzalo, O., Qiu, Y., Jeremias, I., Debose, L., O'Brien, E., Ma, H., Zhou, P., Jacamo, R., Park, E., Coombes, K.R., Zhang, N., Thomas, D.A., O'Brien, S., Kantarjian, H.M., Leverson, J.D., Kornblau, S.M., Andreeff, M., Müschen, M., Zweidler-McKay, P.A., Mulloy, J.C., Letai, A., Milne, T.A., Konopleva, M., (2015). MLL-rearranged acute lymphoblastic leukemias activate BCL-2 through H3K79 methylation and are sensitive to the BCL-2-specific antagonist ABT-199. *Cell Rep.* **13**, 2715–2727.
48. Zhou, C.-C., Xiong, Q.-C., Zhu, X.-X., Du, W., Deng, P., Li, X.-B., Jiang, Y.-Z., Zou, S.-J., Wang, C.-Y., Yuan, Q., (2017). AFF1 and AFF4 differentially regulate the osteogenic differentiation of human MSCs. *Bone Res.* **5**, 1–10.
49. Zhou, J., Ng, Y., Chng, W.-J., (2018). ENL: structure, function, and roles in hematopoiesis and acute myeloid leukemia. *Cell. Mol. Life Sci.* **75**, 3931–3941.
50. Yokoyama, A., Lin, M., Naresh, A., Kitabayashi, I., Cleary, M.L., (2010). A higher-order complex containing AF4 and ENL family proteins with P-TEFb facilitates oncogenic and physiologic MLL-dependent transcription. *Cancer Cell* **17** https://doi.org/10.1016/j.ccr.2009.12.040.
51. Brès, V., Yoh, S.M., Jones, K.A., (2008). The multi-tasking P-TEFb complex. *Curr. Opin. Cell Biol.* **20**, 334–340.
52. Fujinaga, K., Huang, F., Peterlin, B.M., (2023). P-TEFb: the master regulator of transcription elongation. *Mol. Cell* **83**, 393–403.

53. Kim, J., Chu, J., Shen, X., Wang, J., Orkin, S.H., (2008). An extended transcriptional network for pluripotency of embryonic stem cells. *Cell* **132**, 1049–1061.
54. Gallenne, T., Ross, K.N., Visser, N.L., Salony, C.J., Desmet, B.S., Wittner, L.F.A., Wessels, S., Ramaswamy, D.S.P., (2017). Systematic functional perturbations uncover a prognostic genetic network driving human breast cancer. *Oncotarget* **8**, 20572–20587.
55. Lachmann, A., Rizzo, K.A., Bartal, A., Jeon, M., Clarke, D. J.B., Ma'ayan, A., (2023). PrismEXP: gene annotation prediction from stratified gene-gene co-expression matrices. *PeerJ* **11**, e14927
56. HuBMAP Consortium, (2019). The human body at cellular resolution: the NIH Human Biomolecular Atlas Program. *Nature* **574**, 187–192.
57. Aguet, F., Anand, S., Ardlie, K.G., Gabriel, S., Getz, G.A., Graubert, A., Hadley, K., Handsaker, R.E., Huang, K.H., Kashin, S., Li, X., MacArthur, D.G., Meier, S.R., Nedzel, J. L., Nguyen, D.T., Segrè, A.V., Todres, E., Balliu, B., Barbeira, A.N., Battle, A., Bonazzola, R., Brown, A., Brown, C.D., Castel, S.E., Conrad, D.F., Cotter, D.J., Cox, N., Das, S., de Goede, O.M., Dermitzakis, E.T., Einson, J., Engelhardt, B.E., Eskin, E., Eulalio, T.Y., Ferraro, N.M., Flynn, E.D., Fresard, L., Gamazon, E.R., Garrido-Martín, D., Gay, N.R., Gloudemans, M.J., Guigó, R., Hame, A.R., He, Y., Hoffman, P.J., Hormozdiari, F., Hou, L., Im, H.K., Jo, B., Kasela, S., Kellis, M., Kim-Hellmuth, S., Kwong, A., Lappalainen, T., Li, X., Liang, Y., Mangul, S., Mohammadi, P., Montgomery, S.B., Muñoz-Aguirre, M., Nachun, D.C., Nobel, A.B., Oliva, M., Park, Y., Park, Y., Parsana, P., Rao, A.S., Reverter, F., Rouhana, J.M., Sabatti, C., Saha, A., Stephens, M., Stranger, B.E., Strober, B.J., Teran, N.A., Viñuela, A., Wang, G., Wen, X., Wright, F., Wucher, V., Zou, Y., Ferreira, P.G., Li, G., Melé, M., Yeger-Lotem, E., Barcus, M.E., Bradbury, D., Krubit, T., McLean, J.A., Qi, L., Robinson, K., Roche, N.V., Smith, A.M., Sobin, L., Tabor, D.E., Undale, A., Bridge, J., Brigham, L.E., Foster, B.A., Gillard, B.M., Hasz, R., Hunter, M., Johns, C., Johnson, M., Karasik, E., Kopen, G., Leinweber, W.F., McDonald, A., Moser, M.T., Myer, K., Ramsey, K.D., Roe, B., Shad, S., Thomas, J.A., Walters, G., Washington, M., Wheeler, J., Jewell, S.D., Rohrer, D.C., Valley, D.R., Davis, D.A., Mash, D.C., Branton, P.A., Barker, L.K., Gardiner, H.M., Mosavel, M., Siminoff, L.A., Flicek, P., Haeussler, M., Juettemann, T., James Kent, W., Lee, C.M., Powell, C.C., Rosenbloom, K.R., Ruffier, M., Sheppard, D., Taylor, K., Trevanion, S.J., Zerbino, D.R., Abell, N.S., Akey, J., Chen, L., Demanelis, K., Doherty, J.A., Feinberg, A.P., Hansen, K.D., Hickey, P. F., Jasmine, F., Jiang, L., Kaul, R., Kibriya, M.G., Li, J.B., Li, Q., Lin, S., Linder, S.E., Pierce, B.L., Rizzardi, L.F., Skol, A.D., Smith, K.S., Snyder, M., Stamatoyannopoulos, J., Tang, H., Wang, M., Carithers, L.J., Guan, P., Koester, S.E., Roger Little, A., Moore, H.M., Nierras, C.R., Rao, A. K., Vaught, J.B., Volpi, S., (2020). The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science*. https://doi.org/10.1126/science.aaz1776.
58. Li, Y., Dou, Y., Da Veiga Leprevost, F., Geffen, Y., Calinawan, A.P., Aguet, F., Akiyama, Y., Anand, S., Birger, C., Cao, S., Chaudhary, R., Chilappagari, P., Cieslik, M., Colaprico, A., Zhou, D.C., Day, C., Domagalski, M.J., Esai Selvan, M., Fenyö, D., Foltz, S. M., Francis, A., Gonzalez-Robles, T., Gümüş, Z.H., Heiman, D., Holck, M., Hong, R., Hu, Y., Jaehnig, E.J., Ji, J., Jiang, W., Katsnelson, L., Ketchum, K.A., Klein, R.J., Lei, J.T., Liang, W.-W., Liao, Y., Lindgren, C.M., Ma, W., Ma, L., MacCoss, M.J., Martins Rodrigues, F., McKerrow, W., Nguyen, N., Oldroyd, R., Pilozzi, A., Pugliese, P., Reva, B., Rudnick, P., Ruggles, K.V., Rykunov, D., Savage, S.R., Schnaubelt, M., Schraink, T., Shi, Z., Singhal, D., Song, X., Storrs, E., Terekhanova, N.V., Thangudu, R.R., Thiagarajan, M., Wang, L.-B., Wang, J. M., Wang, Y., Wen, B., Wu, Y., Wyczalkowski, M.A., Xin, Y., Yao, L., Yi, X., Zhang, H., Zhang, Q., Zuhl, M., Getz, G., Ding, L., Nesvizhskii, A.I., Wang, P., Robles, A.I., Zhang, B., Payne, S.H., Clinical Proteomic Tumor Analysis Consortium, (2023). Proteogenomic data and resources for pan-cancer analysis. *Cancer Cell* **41**, 1397–1406.
59. Ovaska, K., Laakso, M., Haapa-Paananen, S., Louhimo, R., Chen, P., Aittomäki, V., Valo, E., Núñez-Fontarnau, J., Rantanen, V., Karinen, S., Nousiainen, K., Lahesmaa-Korpinen, A.-M., Miettinen, M., Saarinen, L., Kohonen, P., Wu, J., Westermarck, J., Hautaniemi, S., (2010). Large-scale data integration framework provides a comprehensive view on glioblastoma multiforme. *Genome Med.* **2**, 65.
60. Tsherniak, A., Vazquez, F., Montgomery, P.G., Weir, B.A., Kryukov, G., Cowley, G.S., Gill, S., Harrington, W.F., Pantel, S., Krill-Burger, J.M., Meyers, R.M., Ali, L., Goodale, A., Lee, Y., Jiang, G., Hsiao, J., Gerath, W.F.J., Howell, S., Merkel, E., Ghandi, M., Garraway, L.A., Root, D.E., Golub, T.R., Boehm, J.S., Hahn, W.C., (2017). Defining a cancer dependency map. *Cell* **170**, 564–576. e16.
61. Nusinow, D.P., Szpyt, J., Ghandi, M., Rose, C.M., McDonald 3rd, E.R., Kalocsay, M., Jané-Valbuena, J., Gelfand, E., Schweppe, D.K., Jedrychowski, M., Golji, J., Porter, D.A., Rejtar, T., Wang, Y.K., Kryukov, G.V., Stegmeier, F., Erickson, B.K., Garraway, L.A., Sellers, W. R., Gygi, S.P., (2020). Quantitative proteomics of the cancer cell line encyclopedia. *Cell* **180**, 387–402.e16.
62. Cao, J., O'Day, D.R., Pliner, H.A., Kingsley, P.D., Deng, M., Daza, R.M., Zager, M.A., Aldinger, K.A., Blecher-Gonen, R., Zhang, F., Spielmann, M., Palis, J., Doherty, D., Steemers, F.J., Glass, I.A., Trapnell, C., Shendure, J., (2020). A human cell atlas of fetal gene expression. *Science* **370** https://doi.org/10.1126/science.aba7721.
63. Piñero, J., Ramírez-Anguita, J.M., Saüch-Pitarch, J., Ronzano, F., Centeno, E., Sanz, F., Furlong, L.I., (2019). The DisGeNET knowledge platform for disease genomics: 2019 update. *Nucleic Acids Res* **48**, D845–D855.
64. Abugessaisa, I., Ramilowski, J.A., Lizio, M., Severin, J., Hasegawa, A., Harshbarger, J., Kondo, A., Noguchi, S., Yip, C.W., Ooi, J.L.C., Tagami, M., Hori, F., Agrawal, S., Hon, C.C., Cardon, M., Ikeda, S., Ono, H., Bono, H., Kato, M., Hashimoto, K., Bonetti, A., Kobayashi, N., Shin, J., de Hoon, M., Hayashizaki, Y., Carninci, P., Kawaji, H., Kasukawa, T., (2020). FANTOM enters 20th year: expansion of transcriptomic atlases and functional annotation of non-coding RNAs. *Nucleic Acids Res.* **49**, D892.
65. York, W.S., Mazumder, R., Ranzinger, R., Edwards, N., Kahsay, R., Aoki-Kinoshita, K.F., Campbell, M.P., Cummings, R.D., Feizi, T., Martin, M., Natale, D.A., Packer, N.H., Woods, R.J., Agarwal, G., Arpinar, S., Bhat, S., Blake, J., Castro, L.J.G., Fochtman, B., Gildersleeve, J., Goldman, R., Holmes, X., Jain, V., Kulkarni, S., Mahadik, R., Mehta, A., Mousavi, R., Nakarakommula, S., Navelkar, R., Pattabiraman, N.,

Pierce, M.J., Ross, K., Vasudev, P., Vora, J., Williamson, T., Zhang, W., (2020). GlyGen: computational and informatics resources for glycoscience. *Glycobiology* **30**, 72–73.

66. Ashburner, M., Ball, C.A., Blake, J.A., Botstein, D., Butler, H., Cherry, J.M., Davis, A.P., Dolinski, K., Dwight, S.S., Eppig, J.T., Harris, M.A., Hill, D.P., Issel-Tarver, L., Kasarskis, A., Lewis, S., Matese, J.C., Richardson, J.E., Ringwald, M., Rubin, G.M., Sherlock, G., (2000). Gene ontology: tool for the unification of biology. *Nature Genet.* **25**, 25–29.
67. Sollis, E., Mosaku, A., Abid, A., Buniello, A., Cerezo, M., Gil, L., Groza, T., Güneş, O., Hall, P., Hayhurst, J., Ibrahim, A., Ji, Y., John, S., Lewis, E., MacArthur, J.A.L., McMahon, A., Osumi-Sutherland, D., Panoutsopoulou, K., Pendlington, Z., Ramachandran, S., Stefancsik, R., Stewart, J., Whetzel, P., Wilson, R., Hindorff, L., Cunningham, F., Lambert, S.A., Inouye, M., Parkinson, H., Harris, L.W., (2023). The NHGRI-EBI GWAS catalog: knowledgebase and deposition resource. *Nucleic Acids Res.* **51** https://doi.org/10.1093/nar/gkac1010.
68. Börner, K., Bueckle, A., Herr, B.W., Cross, L.E., Quardokus, E.M., Record, E.G., Ju, Y., Silverstein, J.C., Browne, K.M., Jain, S., Wasserfall, C.H., Jorgensen, M.L., Spraggins, J.M., Patterson, N.H., Weber, G.M., (2022). Tissue registration and exploration user interfaces in support of a human reference atlas. *Commun. Biol.* **5**, 1–9.
69. Su, A.I., Wiltshire, T., Batalov, S., Lapp, H., Ching, K.A., Block, D., Zhang, J., Soden, R., Hayakawa, M., Kreiman, G., Cooke, M.P., Walker, J.R., Hogenesch, J.B., (2004). A gene atlas of the mouse and human protein-encoding transcriptomes. *PNAS* **101** https://doi.org/10.1073/pnas.0400782101.
70. Gargano, M.A., Matentzoglu, N., Coleman, B., Addo-Lartey, E.B., Anagnostopoulos, A.V., Anderton, J., Avillach, P., Bagley, A.M., Bakštein, E., Balhoff, J.P., Baynam, G., Bello, S.M., Berk, M., Bertram, H., Bishop, S., Blau, H., Bodenstein, D.F., Botas, P., Boztug, K., Čady, J., Callahan, T.J., Cameron, R., Carbon, S.J., Castellanos, F., Caufield, J.H., Chan, L.E., Chute, C.G., Cruz-Rojo, J., Dahan-Oliel, N., Davids, J.R., de Dieuleveult, M., de Souza, V., de Vries, B.B.A., de Vries, E., DePaulo, J.R., Derfalvi, B., Dhombres, F., Diaz-Byrd, C., Dingemans, A.J. M., Donadille, B., Duyzend, M., Elfeky, R., Essaid, S., Fabrizzi, C., Fico, G., Firth, H.V., Freudenberg-Hua, Y., Fullerton, J.M., Gabriel, D.L., Gilmour, K., Giordano, J., Goes, F.S., Moses, R.G., Green, I., Griese, M., Groza, T., Gu, W., Guthrie, J., Gyori, B., Hamosh, A., Hanauer, M., Hanušová, K., He, Y.O., Hegde, H., Helbig, I., Holasová, K., Hoyt, C.T., Huang, S., Hurwitz, E., Jacobsen, J.O.B., Jiang, X., Joseph, L., Keramatian, K., King, B., Knoflach, K., Koolen, D.A., Kraus, M.L., Kroll, C., Kusters, M., Ladewig, M.S., Lagorce, D., Lai, M.-C., Lapunzina, P., Laraway, B., Lewis-Smith, D., Li, X., Lucano, C., Majd, M., Marazita, M.L., Martinez-Glez, V., McHenry, T.H., McInnis, M.G., McMurry, J.A., Mihulová, M., Millett, C.E., Mitchell, P. B., Moslerová, V., Narutomi, K., Nematollahi, S., Nevado, J., Nierenberg, A.A., Čajbiková, N.N., Nurnberger Jr, J.I., Ogishima, S., Olson, D., Ortiz, A., Pachajoa, H., Perez de Nanclares, G., Peters, A., Putman, T., Rapp, C.K., Rath, A., Reese, J., Rekerle, L., Roberts, A.M., Roy, S., Sanders, S.J., Schuetz, C., Schulte, E.C., Schulze, T.G., Schwarz, M., Scott, K., Seelow, D., Seitz, B., Shen, Y., Similuk, M.N., Simon, E.S., Singh, B., Smedley, D., Smith, C.L., Smolinsky, J.T., Sperry, S., Stafford, E., Stefancsik, R., Steinhaus, R., Strawbridge, R., Sundaramurthi, J.C., Talapova, P., Tenorio Castano, J.A., Tesner, P., Thomas, R.H., Thurm, A., Turnovec, M., van Gijn, M.E., Vasilevsky, N.A., Vlčková, M., Walden, A., Wang, K., Wapner, R., Ware, J.S., Wiafe, A.A., Wiafe, S.A., Wiggins, L.D., Williams, A.E., Wu, C., Wyrwoll, M.J., Xiong, H., Yalin, N., Yamamoto, Y., Yatham, L.N., Yocum, A.K., Young, A. H., Yüksel, Z., Zandi, P.P., Zankl, A., Zarante, I., Zvolský, M., Toro, S., Carmody, L.C., Harris, N.L., Munoz-Torres, M.C., Danis, D., Mungall, C.J., Köhler, S., Haendel, M.A., Robinson, P.N., (2024). The Human Phenotype Ontology in 2024: phenotypes around the world. *Nucleic Acids Res.* **52** (2024), D1333–D1346.
71. Ursu, O., Holmes, J., Knockel, J., Bologa, C.G., Yang, J.J., Mathias, S.L., Nelson, S.J., Oprea, T.I., (2016). DrugCentral: online drug compendium. *Nucleic Acids Res.* **45**, D932–D939.
72. Pletscher-Frankild, S., Pallejà, A., Tsafou, K., Binder, J.X., Jensen, L.J., (2015). DISEASES: Text mining and data integration of disease–gene associations. *Methods* **74**, 83–89.
73. Kanehisa, M., (2019). Toward understanding the origin and evolution of cellular organisms. *Protein Sci.* **28**, 1947–1951.
74. Peterson, K.A., Murray, S.A., (2022). Progress towards completing the mutant mouse null resource. *Mamm. Genome* **33**, 123–134.
75. Evangelista, J.E., Clarke, D.J.B., Xie, Z., Lachmann, A., Jeon, M., Chen, K., Jagodnik, K.M., Jenkins, S.L., Kuleshov, M.V., Wojciechowicz, M.L., Schürer, S.C., Medvedovic, M., Maa'yan, A., (2022). SigCom LINCS: data and metadata search engine for a million gene expression signatures. *Nucleic Acids Res.* **50**, W697–W709.
76. Sud, M., Fahy, E., Cotter, D., Azam, K., Vadivelu, I., Burant, C., Edison, A., Fiehn, O., Higashi, R., Nair, K.S., Sumner, S., Subramaniam, S., (2016). Metabolomics Workbench: an international repository for metabolomics data and metadata, metabolite standards, protocols, tutorials and training, and analysis tools. *Nucleic Acids Res.* **44** https://doi.org/10.1093/nar/gkv1042.
77. Smith, C.L., Eppig, J.T., (2009). The mammalian phenotype ontology: enabling robust annotation and comparative analysis. *Wiley Interdisc. Rev. Syst. Biol. Med.* **1** https://doi.org/10.1002/wsbm.44.
78. Finn, R.D., Bateman, A., Clements, J., Coggill, P., Eberhardt, R.Y., Eddy, S.R., Heger, A., Hetherington, K., Holm, L., Mistry, J., Sonnhammer, E.L.L., Tate, J., Punta, M., (2013). Pfam: the protein families database. *Nucleic Acids Res.* **42**, D222.
79. Shin, M.G., Pico, A.R., (2023). Using published pathway figures in enrichment analysis and machine learning. *BMC Genomics* **24** https://doi.org/10.1186/s12864-023-09816-1.
80. Mitchell, D.C., Kuljanin, M., Li, J., Van Vranken, J.G., Bulloch, N., Schweppe, D.K., Huttlin, E.L., Gygi, S.P., (2023). A proteome-wide atlas of drug mechanism of action. *Nature Biotechnol.* **41**, 845–857.
81. The Tabula Muris Consortium, (2020). A single cell transcriptomic atlas characterizes aging tissues in the mouse. *Nature* **583**, 590.
82. Tabula Sapiens Consortium, Jones, R.C., Karkanias, J., Krasnow, M.A., Pisco, A.O., Quake, S.R., Salzman, J., Yosef, N., Bulthaup, B., Brown, P., Harper, W., Hemenez,

M., Ponnusamy, R., Salehi, A., Sanagavarapu, B.A., Spallino, E., Aaron, K.A., Concepcion, W., Gardner, J.M., Kelly, B., Neidlinger, N., Wang, Z., Crasta, S., Kolluru, S., Morri, M., Tan, S.Y., Travaglini, K.J., Xu, C., Alcántara-Hernández, M., Almanzar, N., Antony, J., Beyersdorf, B., Burhan, D., Calcuttawala, K., Carter, M.M., Chan, C.K.F., Chang, C.A., Chang, S., Colville, A., Culver, R.N., Cvijović, I., D'Amato, G., Ezran, C., Galdos, F.X., Gillich, A., Goodyer, W.R., Hang, Y., Hayashi, A., Houshdaran, S., Huang, X., Irwin, J.C., Jang, S., Juanico, J.V., Kershner, A. M., Kim, S., Kiss, B., Kong, W., Kumar, M.E., Kuo, A.H., Li, B., Loeb, G.B., Lu, W.-J., Mantri, S., Markovic, M., McAlpine, P.L., de Morree, A., Mrouj, K., Mukherjee, S., Muser, T., Neuhöfer, P., Nguyen, T.D., Perez, K., Puluca, N., Qi, Z., Rao, P., Raquer-McKay, H., Schaum, N., Scott, B., Seddighzadeh, B., Segal, J., Sen, S., Sikandar, S., Spencer, S.P., Steffes, L.C., Subramaniam, V.R., Swarup, A., Swift, M., Van Treuren, W., Trimm, E., Veizades, S., Vijayakumar, S., Vo, K.C., Vorperian, S.K., Wang, W., Weinstein, H.N.W., Winkler, J., Wu, T.T.H., Xie, J., Yung, A.R., Zhang, Y., Detweiler, A.M., Mekonen, H., Neff, N.F., Sit, R.V., Tan, M., Yan, J., Bean, G.R., Charu, V., Forgó, E., Martin, B.A., Ozawa, M.G., Silva, O., Toland, A., Vemuri, V.N.P., Afik, S., Awayan, K., Botvinnik, O.B., Byrne, A., Chen, M., Dehghannasiri, R., Gayoso, A., Granados, A.A., Li, Q., Mahmoudabadi, G., McGeever, A., Olivieri, J.E., Park, M., Ravikumar, N., Stanley, G., Tan, W., Tarashansky, A.J., Vanheusden, R., Wang, P., Wang, S., Xing, G., Dethlefsen, L., Ezran, C., Gillich, A., Hang, Y., Ho, P.-Y., Irwin, J.C., Jang, S., Leylek, R., Liu, S., Maltzman, J.S., Metzger, R.J., Phansalkar, R., Sasagawa, K., Sinha, R., Song, H., Swarup, A., Trimm, E., Veizades, S., Wang, B., Beachy, P.A., Clarke, M.F., Giudice, L.C., Huang, F.W., Huang, K.C., Idoyaga, J., Kim, S.K., Kuo, C.S., Nguyen, P., Rando, T.A., Red-Horse, K., Reiter, J., Relman, D.A., Sonnenburg, J.L., Wu, A., Wu, S. M., Wyss-Coray, T., (2022). The Tabula Sapiens: a multiple-organ, single-cell transcriptomic atlas of humans. *Science* **376**, eabl4896

83. Johnson, J.L., Yaron, T.M., Huntsman, E.M., Kerelsky, A., Song, J., Regev, A., Lin, T.-Y., Liberatore, K., Cizin, D.M., Cohen, B.M., Vasan, N., Ma, Y., Krismer, K., Robles, J.T., van de Kooij, B., van Vlimmeren, A.E., Andrée-Busch, N., Käufer, N.F., Dorovkov, M.V., Ryazanov, A.G., Takagi, Y., Kastenhuber, E.R., Goncalves, M.D., Hopkins, B.D., Elemento, O., Taatjes, D.J., Maucuer, A., Yamashita, A., Degterev, A., Uduman, M., Lu, J., Landry, S.D., Zhang, B., Cossentino, I., Linding, R., Blenis, J., Hornbeck, P.V., Turk, B.E., Yaffe, M.B., Cantley, L.C., (2023). An atlas of substrate specificities for the human serine/threonine kinome. *Nature* **613**, 759–766.
84. Han, H., Cho, J.-W., Lee, S., Yun, A., Kim, H., Bae, D., Yang, S., Kim, C.Y., Lee, M., Kim, E., Lee, S., Kang, B., Jeong, D., Kim, Y., Jeon, H.-N., Jung, H., Nam, S., Chung, M., Kim, J.-H., Lee, I., (2018). TRRUST v2: an expanded reference database of human and mouse transcriptional regulatory interactions. *Nucleic Acids Res.* **46**, D380–D386.